

Parul Institute of Engineering & Technology

Engineering Department

ASSIGNMENT-3

Subject Name: DIGITAL ELECTRONICS

Subject Code: 303105220

(Chapter-3-Combinational Logic)

1. Design the Combinational Circuits for Binary to Gray Code Conversion.
2. Explain Design Procedure for Combinational Circuit & Difference between Combinational Circuit & Sequential Circuit.
3. Construct 4*16 Decoder with help of 2*4 Decoder.
4. Discuss 4 bit BCD Adder in Detail.
5. Explain 4 bit Magnitude Comparator.
6. Explain common cathode types seven segments displays.
7. Design combinational circuits for a full adder.
8. Design a full-adder with two half-adders and an OR gate
9. Design a combinational circuit for a full subtractor.
10. Design combinational circuits for a half adder.
11. Design a combinational circuit for a half subtractor.
12. Design a combinational circuit whose input is a four bit number and whose Output is the 2's complement of the input number.
13. Draw symbol and truth table for four input EX-OR and EX-NOR gate. Explain NAND and NOR as an universal gate.
14. Design BCD to Excess-3 code converter using minimum number of NAND gates.
15. Simplify Boolean function $F(w,x,y,z) = \sum (0,1,2,4,5,6,8,9,12,13,14)$ using K-map